

Java Question Bank

Arrays

1. Write a Java program to find the largest and smallest elements in an integer array.
2. Input an array of integers and print all prime numbers from it.
3. Write a program to reverse an array without using another array.
4. Find the frequency of each element in an array.
5. Write a program to sort an array in ascending order using the Bubble Sort algorithm.
6. Input marks of 10 students into an array and calculate the average score.
7. What will be the output?

```
int[] arr = {2, 4, 6, 8};  
System.out.println(arr[1] + arr[3]);
```

Loops

1. Write a program to print the multiplication table of any number entered by the user.
2. Display all numbers between 1 and 200 that are divisible by 7 using a loop.
3. Write a program to find the sum of digits of a number using a while loop.
4. Print the following pattern (n=4):
1
12
123
1234
5. What will the following code output?

```
for (int i = 1; i <= 5; i++) {  
    if (i == 3) continue;  
    System.out.print(i + " ");  
}
```

6. Write a program using a nested loop to print a square matrix of size 4×4 filled with *.
7. Write a program to check whether a number is palindrome using loops.

Recursion

1. Write a recursive function to calculate factorial of a given number.
2. Write a recursive method to find the nth Fibonacci number.
3. What will be the output?

```
int fun(int n) {  
    if (n == 0) return 0;  
    return n + fun(n - 1);  
}  
System.out.println(fun(3));
```
4. Write a recursive function to find the sum of digits of a number.
5. Explain what a base case is in recursion. Why is it necessary?
6. Write a recursive function to reverse a string.
7. Differentiate between recursion and iteration with examples.

Strings

1. Write a Java program to check whether a string is a palindrome.
2. Input a string and count the number of vowels and consonants.
3. Write a program to find the frequency of each character in a string.
4. Write a program to reverse each word in a sentence (e.g., 'Hello World' → 'olleH dlroW').
5. What is the output of the following code?

```
String s = "Java Programming";  
System.out.println(s.substring(5, 11));
```

6. Write a program to remove all spaces from a string.
7. Explain the difference between equals() and == in Java strings.

Conditional Statements

1. Write a Java program to check whether a given number is positive, negative, or zero.
2. Input marks of a student and print grade according to rules: $\geq 90 \rightarrow A$, 75–89 $\rightarrow B$, 50–74 $\rightarrow C$, $< 50 \rightarrow \text{Fail}$.
3. Write a program to check whether a given year is a leap year.
4. Use a switch-case to display the day of the week based on user input (1 = Sunday, ... 7 = Saturday).
5. What is the output?

```
int x = 10, y = 20;  
if (x > y)  
    System.out.println("X is greater");  
else  
    System.out.println("Y is greater");
```
6. Write a Java program to find the maximum of three numbers using if-else.
7. Differentiate between if-else if ladder and switch-case with an example.